**2 (a) (i)** Gravitational force provides centripetal force. Same gravitational force acting on star and planet.

$$F_G = m_p r_p \omega^2 = m_s r_s \omega^2$$ C1

$$2.0 \times 10^{27} \left(r_p\right) = 6.0 \times 10^{28} \left(r_s\right)$$

$$\frac{r_s}{r_p} = \frac{2.0 \times 10^{27}}{6.0 \times 10^{28}} = \frac{1}{30}$$

$$r_s = \frac{1}{31} \times 5.2 \times 10^9 = 1.68 \times 10^8 \text{ m}$$ A1

**(a) (ii)**

$$F_G = F_C$$

$$\frac{G m_s m_p}{d^2} = m_s r_s \omega^2$$

$$\frac{G m_s m_p}{d^2} = m_s r_s \left(\frac{2\pi}{T}\right)^2$$ C1

$$\frac{6.67 \times 10^{-11}\left(2.0 \times 10^{27}\right)}{\left(5.2 \times 10^9\right)^2} = \left(1.68 \times 10^8\right)\left(\frac{2\pi}{T}\right)^2$$

$$T = 1.16 \times 10^6 \text{ s}$$ A1

**(a) (iii)** Planet moves in front of star periodically. A1

**(b) (i)** Between starting height $6.40 \times 10^6$ m & ending height $7.225 \times 10^6$ m

$$g = (-)\frac{d\phi}{dr}$$

$$\Delta\phi = \int g\ dr$$ C1

$= \text{area under g-r graph}$ M1

$= 6.74 \times 10^6 \text{ to } 7.61 \times 10^6 \text{ J kg}^{-1}$

M1

$$\Delta U = m\Delta\phi = 2300\ \Delta\phi$$

$= 1.55 \times 10^{10}$ to $1.75 \times 10^{10}$ J

OR

$$g = G\frac{M}{r^2} \text{ and } \phi = -G\frac{M}{r} \Rightarrow \phi = -gr$$ C1

$$\Delta U = m\Delta\phi = m(\phi_f - \phi_i) = 2300(g_i r_i - g_f r_f)$$ M1

$$= 2300(-9.75 \times 6.4 \times 10^6 + 7.625 \times 7.225 \times 10^6)$$ M1

$\approx 1.60 \times 10^{10}$ J

**(b) (ii)**

$$\Delta U = U_f - U_i = -\frac{GM_E m_s}{r_f} - \left(-\frac{GM_E m_s}{r_i}\right)$$

$$1.6 \times 10^{10} = 6.67 \times 10^{-11} M_E\,(2300)\left(\frac{1}{6.40 \times 10^6} - \frac{1}{7.225 \times 10^6}\right)$$ C1

$$M_E = 5.85 \times 10^{24} \text{ kg}$$ A1

**(b) (iii)** Advantages to low polar orbit:

- High(er) resolution imaging/clearer images A1
- Image more of the planet as the Earth spins underneath the satellite

Geostationary:

- **Remain at the same position in the sky so satellite dishes can keep locked on to signal/no steerable dishes needed/send & receive signals all the times/continuous/uninterrupted/stable signals** A1
- Higher orbit means greater coverage of the signals

**3** **(a)** ... $^{-1}$ A1